Phase-Associative Memory: Sequence Modeling in Complex Hilbert Space

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We present Phase-Associative Memory (PAM), a recurrent sequence model in which all representations are complex-valued, associations accumulate in a matrix state $S_t \in \mathbb{C}^{d\times d}$ via outer products, and retrieval operates through the conjugate inner product $K_t^* \cdot Q_t/\sqrt{d}$. At ∼100M parameters on WikiText-103, PAM reaches validation perplexity 30.0, within ∼10% of a matched transformer (27.1) trained under identical conditions, despite 4× arithmetic overhead from complex computation and no custom kernels. We trace the experimental path from vector-state models, where holographic binding fails due to the $O(1/\sqrt{n})$ capacity degradation of superposed associations, to the matrix state that resolves it. The competitiveness of an architecture whose native operations are complex-valued superposition and conjugate retrieval is consistent with recent empirical evidence that semantic interpretation in both humans and large language models exhibits non-classical contextuality, and we discuss what this implies for the choice of computational formalism in language modeling.

INTRODUCTION

The assumption that a system's constituents can be analyzed independently of one another and of the conditions under which they are observed has been a foundational premise of empirical science since the seventeenth century [1, 2]. Scholastic metaphysics, notably Aquinas [3], treated nature as composed of distinct substances knowable in isolation [4]; early modern figures from Galileo [5] and Bacon (*Novum Organum* [6]) through Descartes [7] to Newton's *Principia* [8] sharpened a picture in which states and causes admit description without essential reference to the observer.

For more than two centuries classical physics reinforced separability and determinism as features of the world. Quantum mechanics challenged that picture: Heisenberg preserved pre-existing properties disturbed by measurement, whereas Bohr held quantities indeterminate prior to measurement. Einstein, Podolsky, and Rosen argued for definite properties of separated systems [9]; separability was central [10, 11]. Bell's theorem [12, 13] and experiments [14, 15] favored Bohr [16]: no pre-existing, context-independent values reproduce the correlations. The Kochen–Specker theorem [17] extended contextuality to single systems. Later tests closed loopholes [18–21]; quantum information [22–24] showed inseparability as a resource, with the Tsirelson bound [25] quantifying quantum-over-classical advantage.

The study of language has developed under the same separability assumptions, though the connection is rarely made explicit. The principle of compositionality, which holds that the meaning of a complex expression is determined entirely by the meanings of its parts and the rules by which they combine [26, 27], treats semantic content as a property of linguistic constituents that can be analyzed independently of the interpreter and the context of interpretation. Whether this principle is adequate as a foundation for the study of meaning has been contested on philosophical grounds for more than a century [28–30], but the computational study of language has largely proceeded as if it were settled. Zellig Harris showed that the distributional properties of words in a corpus could serve as a proxy for their semantic relationships [31], and this insight carried through the twentieth century into latent semantic analysis, word embeddings [32, 33], and ultimately large language models [34–36]. At each stage, the underlying computational assumption has been the same: words have meanings that can be represented as fixed points in a real-valued vector space, and the task of a model is to learn where those points are and how they compose. The transformer architecture [37, 38] is the most successful embodiment of this program.

Transformer-based large language models have largely succeeded in passing the well-established benchmarks of artificial intelligence, including conversational assessments that would have been considered definitive demonstrations of language understanding a decade ago [36]. However, their adoption in domains that require guaranteed reliability has been hindered by persistent difficulties, most prominently hallucination [39, 40] and susceptibility to prompt injection [41, 42], which have resisted solution despite substantial engineering effort across architectures and scales. The improvements in capability that once accompanied increases in model size and training data [43, 44] have plateaued [45, 46], and the frontier of the field has shifted toward test-time compute [47], chain-of-thought reasoning, and agentic iteration as strategies for navigating the space of possible responses rather than producing the correct one directly. This shift is consistent with the observation that the informational burden of disambiguating a semantic expression grows superlinearly with its complexity [48], making the recovery of a single intended meaning from an expression of even moderate depth an intractable prob-

lem in the sense of relevance realization [49, 50]: the system cannot determine what is relevant from the input alone and must instead explore. Efforts to understand these systems through mechanistic interpretability, which attempts to decompose the internal representations of neural networks into individually meaningful components, have encountered difficulties that appear to be structural rather than merely technical. Sparse autoencoders trained on frontier models lose approximately 90% of the model's capability when their reconstructions replace the original activations [51], extracted features have been shown to be "neither selective nor independent" when used for steering [52], and a recent review with twenty-nine co-authors described the field's foundational concepts as "not yet established" and its status as "pre-paradigmatic" [53]. Theoretical work has demonstrated an exponential gap between the complexity of representing features in superposition and computing with them [54], and sparse autoencoders have been proven to fail to recover ground-truth features except under conditions of extreme sparsity [55]. The difficulty of producing reproducible, meaningful decompositions of neural network representations mirrors the experience of cognitive neuroscience, where decades of functional neuroimaging have shown that localized brain-behavior associations require sample sizes orders of magnitude larger than most studies have used [56], that seventy independent teams analyzing the same fMRI dataset reach substantially different conclusions about which brain regions are involved [57], and that inferring cognitive processes from regional activation is logically unreliable because brain regions participate in many functions simultaneously [58, 59]. In both cases, the assumption that the system can be understood by decomposing it into separable, localizable components has produced results that do not replicate.

In physics, the distinction between a decomposition that has not yet been found and one that cannot exist in principle, because the underlying properties are fundamentally indeterminate prior to measurement, is precisely what Bell's theorem was designed to settle, and the same framework can be applied to semantic interpretation. When the CHSH test is applied to human semantic judgments, the correlations between interpretations produced under different contextual framings violate the classical bound [60–65], and when the same tests are applied to large language models trained on text that human cognition produced, the violations persist across four orders of magnitude in parameter count, with the distributional character of the contextuality orthogonal to every standard benchmark tested [48, 66]. Sheaf-theoretic analysis of BERT's internal representations has identified over 77,000 instances of contextuality at the level of the embeddings themselves [67, 68]. The non-separability is not confined to the behavioral outputs of these systems; it is present in the geometry of their learned representations [69, 70]. Reconsidered under the premise that meaning is indeterminate prior to the act of interpretation and that natural language is semantically degenerate [48], it necessarily follows that hallucinations and jailbreaks are not anomalies to be eliminated but commonplace consequences of a system that interprets rather than retrieves [66]. If the correlational structure of language is genuinely non-classical, the natural mathematical framework for describing it is the same one that was developed for quantum mechanics: a complex Hilbert space in which states carry phase, similarities are computed through the conjugate inner product, and interference between components is an intrinsic property of the algebra rather than a behavior that must be learned. Large language models built on real-valued representations and softmax attention may functionally replicate this structure, but they do so in the way that any classical simulation of a quantum system does: by using enough parameters to project the complex-valued correlations onto a real-valued space, at a cost in capacity and efficiency that grows with the complexity of the structure being represented.

The representation of signals in complex form has a long history in engineering and physics. Gabor [71] introduced the analytic signal in his theory of communication, and Oppenheim and Lim [72] demonstrated that phase carries more structural information than magnitude in both images and audio [72]. The geometric phase discovered independently by Pancharatnam in optics [73] and Berry in quantum mechanics [74] showed that phase relationships encode topological properties of the space traversed by a system, information that is lost entirely when the representation is projected onto real-valued magnitudes. Complex-valued neural networks have been developed along these lines for decades [75–79], and the holographic reduced representations introduced by Plate [80] demonstrated that complex multiplication and conjugation provide a natural algebra for binding and retrieving associations [81, 82]. Danihelka *et al.* [83] incorporated this algebra into an LSTM with complex-valued cell states, and Ramsauer *et al.* [84] showed that the mathematical structure underlying softmax attention is a modern Hopfield network [85, 86] whose linear variant is the fast weight programmer [87, 88]. None of these efforts, however, has produced a complete language model that operates in complex space from embedding through retrieval to output at a scale where comparison with conventional architectures is meaningful.

Separately, the development of efficient alternatives to the transformer's attention mechanism has produced a body of work that provides the architectural scaffolding for such a model. The transformer [37] computes attention as a softmax-normalized dot product between real-valued projections of the input, an operation that is powerful but quadratic in sequence length and requires a key-value cache that grows linearly during inference.

Removing softmax yields a recurrence with matrix state $S_t = S_{t-1} + V_t K_t^\top$ [89], which Schlag *et al.* [88] showed is equivalent to the fast weight programmer introduced by Schmidhuber [87], an associative memory that accumulates associations via outer products and retrieves via matrix-vector product. Subsequent work has refined this structure in various ways: RetNet [90] adds exponential decay, GLA [91] introduces data-dependent gating, DeltaNet [92] replaces additive accumulation with a delta rule, and GateLoop [93] uses complex-valued gates. From the state-space model side, the Linear Recurrent Unit [94] established the importance of complex-valued diagonal recurrences for stable long-range modeling, Mamba [95] introduced input-dependent selection, Griffin [96] validated gated linear recurrence at scale, and Mamba-2 [97] proved the formal equivalence between structured SSMs and linear attention. From the LSTM lineage, mLSTM [98] independently arrives at the same matrix-state recurrence, and RWKV [99, 100] has demonstrated this family at up to 14B parameters. With few exceptions, these models operate in real-valued space. Ramsauer *et al.* [84] showed that softmax attention implements a modern Hopfield network, and the linear variant of this associative memory is precisely the fast weight programmer that the matrix-state models generalize.

The tradition of operational quantum logic, beginning with Birkhoff and von Neumann's observation that the propositions of quantum mechanics form a non-Boolean lattice [101] and developed through the work of Mackey, Piron [102], and Foulis and Randall [103, 104], spent half a century establishing that any system whose observables are contextual requires a non-Boolean algebraic structure, and that the natural home for this structure is a complex Hilbert space with the conjugate inner product. The point is not that such systems are doing quantum physics but that the algebra of complex-valued superposition and interference is the correct formalism for contextual measurement regardless of substrate. Transformers already capture the non-classical correlational structure of language, as the Bell violations demonstrate, but they do so by projecting it onto real-valued space using enough parameters to approximate the complex-valued correlations. Phase-Associative Memory (PAM) takes the matrix-state recurrence shared by the lineages described above and moves it into the space that operational quantum logic identifies as native to contextual systems. The state, keys, values, and queries are all complex-valued, and retrieval uses the conjugate inner product $K^* \cdot Q$ rather than the standard dot product, so that the selectivity of retrieval depends on the phase alignment between stored and queried representations.

The architecture emerged through a series of experiments in which each failure was informative. Early versions introduced tokens in complex phase space but destroyed phase information by passing representations through real-valued nonlinearities; correcting this with phase-preserving primitives materially improved results. A subsequent attempt to inject holographic key–value bindings into a vector-state SSM caused a regression in perplexity, because multiple bindings superposed in a single d-dimensional vector interfere destructively with the classical $O(1/\sqrt{n})$ capacity degradation [80]. PAM resolves this by upgrading the state from $\mathbb{C}^d$ to $\mathbb{C}^{d\times d}$, providing $O(d^2)$ associative capacity per head. The reported configuration interleaves channel mixing and PAM in each of 16 blocks with complex rotary position embeddings [105] on queries and keys, and admits a dual computational form that is $O(T^2)$ for parallel training and $O(1)$ per token for recurrent inference with no KV cache.

At ∼100M parameters on WikiText-103, PAM reaches validation perplexity 30.0 after 10 epochs on a single RTX 4090. A matched transformer trained under identical conditions reaches 27.1. Both are single training runs, and the ∼10% gap should be interpreted accordingly; multi-seed validation is in progress. The gap was achieved with a first-generation pure PyTorch implementation, no custom CUDA kernels, and 4× the arithmetic overhead of real-valued computation, which means the interesting quantity is not the absolute gap but the gap relative to the overhead: PAM pays 4× in arithmetic and loses only ∼10% in perplexity, suggesting the complex formalism is capturing structure efficiently even before the implementation is optimized.

METHOD

The model consists of a complex-valued embedding layer, 16 identical blocks, and a tied complex output head. Each block applies channel mixing via a ComplexGatedUnit (CGU) followed by sequence mixing via a Phase-Associative Memory (PAM) layer, both with residual connections and learned scaling. All operations in the main signal path are complex-valued and phase-preserving; gates and decay parameters use real-valued projections over magnitude features, but the primary data path never converts complex representations to real-valued intermediate forms.

Complex quantities are represented as tensors with shape $[\ldots, d, 2]$, implementing $\mathbb{C}^d$ in split-real form. The complex linear map, given weight matrices $W_r, W_i \in \mathbb{R}^{m\times n}$, computes $y_r = W_r x_r - W_i x_i$ and $y_i = W_i x_r + W_r x_i$. The activation function is modReLU, $\mathrm{modReLU}(z) = \mathrm{ReLU}(|z| + b) \cdot z/|z|$ with learned bias b, which thresholds magnitude while leaving phase untouched. Normalization is RMS normalization applied to magnitudes with phase preserved: $\mathrm{ComplexNorm}(z) = s \cdot (|z|/\,\mathrm{RMS}(|z|)) \cdot z/|z|$ with learned scale s. The channel mixing layer (CGU) is a SwiGLU-style gating block

in complex space:

$$\mathrm{CGU}(z) = W_{\mathrm{down}}\big(\mathrm{gate}_{\mathrm{phase}} \odot \mathrm{modReLU}(W_{\mathrm{up}} z) \cdot \sigma(|W_g z|)\big) \tag{1}$$

where the gate magnitude $\sigma(|W_g z|)$ controls how much signal passes and the gate phase controls what rotation is applied. Each of 16 blocks applies CGU then PAM with residual connections and learned scaling:

$$\tilde{z}^{(l)} = z^{(l-1)} + \alpha^{(l)}_{\mathrm{CGU}} \cdot \mathrm{CGU}_l(\mathrm{ComplexNorm}(z^{(l-1)})), \tag{2}$$

$$z^{(l)} = \tilde{z}^{(l)} + \alpha^{(l)}_{\mathrm{PAM}} \cdot \mathrm{PAM}_l(\mathrm{ComplexNorm}(\tilde{z}^{(l)})) \tag{3}$$

where $\alpha^{(l)}_{\mathrm{CGU}}$ is initialized to 1.0 and $\alpha^{(l)}_{\mathrm{PAM}}$ to 0.1. Logits are computed via a tied complex inner product with the embedding table: $\mathrm{logits} = z_{\mathrm{out},r} \cdot E_r^\top + z_{\mathrm{out},i} \cdot E_i^\top$.

PAM replaces both the recurrent backbone and the attention mechanism with a single module whose operations correspond directly to the quantum semantic framework described in [48]. In that framework, a semantic expression S_E is represented as a state vector $|\psi_{S_E}\rangle = \sum_i c_i |e_i\rangle$ in a complex Hilbert space, where the complex coefficients c_i carry phase information with no classical analogue, and interpretation is the application of a Hermitian operator whose eigenstates represent possible meanings. PAM implements this structure computationally: tokens are embedded as complex vectors, associations between them are accumulated in a complex matrix state via outer products, and retrieval is the projection of a query onto the accumulated state through the conjugate inner product, the same operation that computes $P(m_i) = |\langle e_i|\psi_{S_E}\rangle|^2$ in the quantum semantic framework.

Each PAM head h maintains a complex matrix state $S_t^{(h)} \in \mathbb{C}^{d\times d}$, where d is the head dimension, so that the total state capacity across H heads is $H \times d^2$ complex values per layer ($6 \times 64^2 = 24{,}576$ in our configuration). The input $x_t \in \mathbb{C}^D$ is projected into queries, keys, and values via a single complex linear map:

$$[Q_t; K_t; V_t] = W_{\mathrm{QKV}} x_t \quad \Rightarrow \quad Q_t, K_t, V_t \in \mathbb{C}^{H\times d}. \tag{4}$$

Complex rotary position embeddings [105] are applied to Q and K by multiplying each element by a precomputed unit-magnitude factor $e^{im\theta}$, encoding absolute position in phase while leaving magnitudes unchanged; in the conjugate product $K_i^* \cdot \tilde{Q}_t$ the dependence on position difference $(m-n)$ yields relative position structure. Retrieval uses the scaled query $\tilde{Q}_t := Q_t/\sqrt{d}$.

The decay rate γ_t controls how quickly the state forgets and is computed from the input as $\gamma_t = \exp(-\,\mathrm{softplus}(W_{dt} \cdot \mathrm{concat}(x_{t,r}, x_{t,i}) + b_{dt}))$, where b_{dt} is initialized to -4.0 for slow initial decay. A learned protect gate $p_t = \sigma(W_p \cdot |x_t| + b_p)$ with $b_p = -3.0$ modifies the effective decay:

$$\gamma_t = e^{-dt_t} \cdot (1 - p_t) + p_t, \qquad V_t' = V_t \cdot (1 - p_t). \tag{5}$$

TABLE I. medium-pam-v3 configuration (reported).

Parameter	Value
Complex dimension (D)	384
Blocks	16 (each: CGU → PAM)
Interleaved CGU + PAM	yes
CGU expansion factor	3
PAM heads (H)	6
PAM head dimension (d)	64
Gated State Protection	enabled
Complex RoPE on Q, K	yes
QK phase normalization	off (see)
Sequence length	2048
Total parameters	∼100.4M

When $p_t \to 1$ the state is frozen and new values are suppressed; when $p_t \to 0$ the decay proceeds normally. The state then evolves as:

$$S_t = \gamma_t \cdot S_{t-1} + V_t' \otimes K_t^* \tag{6}$$

where $\otimes$ denotes complex outer product and K_t^* is the complex conjugate of the key. Retrieval computes $Y_t = S_t \tilde{Q}_t$, which expands to:

$$Y_t = \sum_{i\le t} \left(\prod_{j=i+1}^{t} \gamma_j \right) \left(K_i^* \cdot \tilde{Q}_t\right) V_i'. \tag{7}$$

The conjugate inner product $K_i^* \cdot \tilde{Q}_t$ determines retrieval strength through phase alignment: associations whose keys are phase-coherent with the query are retrieved strongly while phase-incoherent associations are suppressed, without softmax normalization.

During training, the recurrence is computed in $O(T^2)$ time by forming a decay matrix $D \in \mathbb{R}^{T\times T}$ with $\log D[t,i] = \sum_{j=i+1}^{t} \log \gamma_j$ via cumulative sums, applying a causal mask, computing the complex score matrix $W = \tilde{Q} K^{*\top}$, and obtaining the output as $Y = (W \odot D) \cdot V'$. This is mathematically equivalent to the recurrence but parallelizes across the sequence dimension. During autoregressive generation, each token requires $O(Hd^2)$ work per layer, and the state $S \in \mathbb{C}^{H\times d\times d}$ is fixed-size and does not grow with sequence length.

We train and evaluate on WikiText-103 [106], approximately 103 million tokens of Wikipedia text tokenized with the GPT-2 BPE tokenizer (vocabulary size 50,257). The primary reported model configuration is listed in Table I. See I.

Training hyperparameters are listed in Table II.

Generation samples are logged every 5,000 steps using temperature 1.0, top-k 50, top-p 0.9, and repetition penalty 1.2.

TABLE II. Training hyperparameters (medium-pam-v3).

Parameter	Value
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	0.01
Warmup steps	1000
LR schedule	warmup + cosine decay
Batch size	3
Epochs	10
Gradient clipping	1.0
Precision	automatic mixed precision (bf16)
Hardware	single NVIDIA RTX 4090 (24GB)
Compilation	torch.compile (default mode)
Initialization	orthogonal (complex linear maps)

TABLE III. medium-pam-v3: training and validation perplexity by epoch (WikiText-103).

Epoch	Train PPL	Val PPL
1	123.86	57.94
2	53.87	43.83
3	44.88	38.69
4	40.39	35.88
5	37.42	33.82
6	35.13	32.25
7	33.26	31.22
8	31.78	30.40
9	30.66	30.01
10	30.02	30.0

RESULTS

The interleaved configuration with complex RoPE reaches validation perplexity 30.0 after 10 epochs on WikiText-103 (single run, RTX 4090). Fig. 1 and Table III summarize training progress.

Total wall time 50,714 s (∼14.1 h). Throughput ∼23k tokens/second average.

We briefly trained a variant with per-element unit normalization of Q and K before the conjugate inner product. Validation loss decreased, but generation collapsed into severe lexical repetition by mid-training and the run was stopped during epoch 5, indicating that both magnitude and phase must be free to vary for the conjugate inner product to function as a retrieval mechanism without softmax. An earlier sequential configuration (16 CGU layers followed by 16 PAM layers, no RoPE, lower learning rate) achieved 38.95 validation PPL, and a hybrid that adds sparse windowed attention every 4th block achieved 30.01, marginally worse than pure PAM, indicating that interleaving channel and sequence mixing is important and that supplemental attention provides no benefit at this scale.

To provide a rigorous comparison, we trained a standard transformer with ∼100.3M parameters on the same

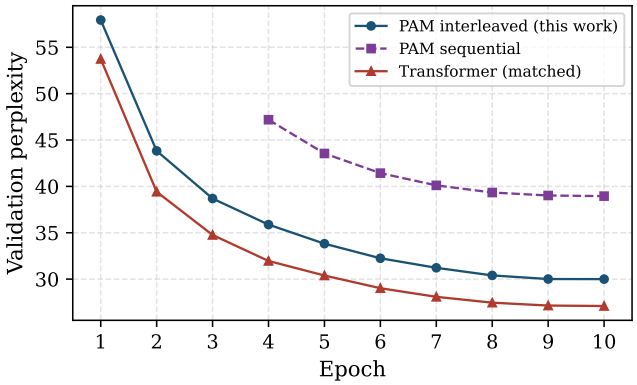

FIG. 1. Validation perplexity vs. epoch for the interleaved PAM configuration, the sequential PAM configuration, and the matched transformer baseline, all trained on WikiText-103.

TABLE IV. Matched comparison on WikiText-103.

Model	Params	Val PPL	tok/s
Transformer (ours)	100.3M	27.1	96k
PAM (ours)	100.4M	30.0	23k

WikiText-103 data, tokenizer (GPT-2 BPE), sequence length (2048), batch size (3), learning rate (1×10^{-4}), warmup (1000 steps), and hardware (single RTX 4090). The transformer uses $d_{\text{model}} = 672$, 12 layers, 12 attention heads, and $d_{\text{ff}} = 2688$.

The ∼10% gap in perplexity is accompanied by a ∼4× gap in training throughput (23k vs. 96k tok/s), which is expected given that complex linear maps require four real matrix multiplies and the PAM implementation uses no custom CUDA kernels. For external reference, GPT-2 [107] at 124M parameters achieves ∼31 validation PPL on WikiText-103, though differences in evaluation protocol make direct comparison unreliable.

The transformer converges faster and maintains a consistent advantage throughout training (Table V), though the gap narrows in later epochs. The two models see the same total tokens. Despite the perplexity gap, the structural differences between the architectures are substantial: the transformer requires $O(T)$ work per token during inference because of the KV cache, whereas PAM's state is fixed at 49,152 floats per layer regardless of sequence length, giving it $O(1)$ per-token cost with per-layer work $O(Hd^2)$. At $T = 2048$ with the baseline's configuration, the transformer's KV cache per layer is ∼56× larger than PAM's state, and this ratio grows linearly with context length.

At epoch 10, given the prompt "In 1923, the University of," the model generates: "In 1923, the University of Illinois at Urbana @-@ Urdu said it was 'an easy choice to do something in its own right.' The university also claimed the first students from Wisconsin had to be replaced by

TABLE V. Epoch-by-epoch validation PPL: matched transformer vs. medium-pam-v3.

	Transformer (ours)		PAM (ours)	
Epoch	Train PPL	Val PPL	Train PPL	Val PPL
1	137.77	53.74	123.86	57.94
2	52.81	39.42	53.87	43.83
3	42.74	34.76	44.88	38.69
4	38.27	31.96	40.39	35.88
5	35.54	30.39	37.42	33.82
6	33.41	29.02	35.13	32.25
7	31.65	28.09	33.26	31.22
8	30.28	27.46	31.78	30.40
9	29.29	27.15	30.66	30.01
10	28.75	27.1	30.02	30.0

a more 'good student' due to a lack of funds." The text is grammatically coherent and shows structural awareness of dates, proper nouns, and institutional language, with 3-gram repetition rate 0.034, 4-gram repetition rate 0.011, and unique token ratio 0.703. Factual accuracy is not reliable at this scale.

DISCUSSION

Schlag *et al.* [88] showed that linear attention without softmax is equivalent to a fast weight programmer [87] in which outer products of keys and values are accumulated into a matrix that queries retrieve from via matrix-vector product. Subsequent work has refined this structure with exponential decay [90], data-dependent gating [91], delta-rule updates [92], and complex-valued gates [93], while from the LSTM lineage Beck *et al.* [98] arrived at the same recurrence independently. PAM belongs to this family, but the substitution of the conjugate inner product for the standard dot product introduces a qualitative difference that is worth examining in the context of what is known about the limitations of linear attention.

Arora *et al.* [108] demonstrated that linear attention models struggle with associative recall, the ability to retrieve a specific stored value given a matching key, because the non-negative nature of real-valued inner products means that all stored associations contribute positively to the retrieval, diluting the target signal. Ramsauer *et al.* [84] showed that softmax attention avoids this problem because it implements a modern Hopfield network [85, 86] in which the exponential nonlinearity sharpens retrieval to a near-one-hot selection. PAM's conjugate inner product provides a different mechanism for addressing the same problem: because the complex dot product $K_i^* \cdot \tilde{Q}_t$ can be negative or imaginary depending on the phase relationship between the stored key and the query, associations that are phase-incoherent with the query are actively suppressed rather than merely downweighted. Whether this destructive interference is as effective as the exponential sharpening of softmax at recovering specific associations from a large store is a question that passkey retrieval experiments at long context lengths would address directly.

Danihelka *et al.* [83] previously incorporated complex-valued holographic representations into an LSTM and found improvements on tasks requiring compositional binding and retrieval, consistent with the classical result of Oppenheim and Lim [72] that phase carries more structural information than magnitude. The connection between phase and representational capacity has a deeper grounding in the geometric phase of Berry [74] and Pancharatnam [73], which showed that phase relationships encode topological properties of the trajectory a system follows through its state space. In the context of PAM, this suggests that the phase of the complex state may encode not only the content of stored associations but something about the path through which they were accumulated, a form of sequential information that real-valued states cannot represent without additional mechanisms. The QK phase normalization ablation described in , in which removing the magnitude degree of freedom from queries and keys caused generation to collapse into repetition, provides indirect evidence that both magnitude and phase are carrying distinct and necessary information in the retrieval process.

Lo *et al.* [67] used a sheaf-theoretic framework [68] to identify over 77,000 instances of contextuality in BERT's internal representations, demonstrating that the non-separability observed behaviorally in Bell violation experiments [48, 60, 64, 66] is also present in the geometry of the embeddings themselves. This finding connects to the difficulties documented in the mechanistic interpretability literature, where sparse autoencoders trained on frontier models lose approximately 90% of the model's capability when their reconstructions replace the original activations [51], extracted features are neither selective nor independent when used for steering [52], and there is a proven exponential gap between representing features in superposition and computing with them [54]. Williams *et al.* [69] have argued that these difficulties may reflect conceptual limitations of the decomposition framework rather than technical ones. If the representations are fundamentally non-separable, as the Bell violations and the sheaf-theoretic analysis suggest, then an architecture that operates natively in complex Hilbert space may not merely provide an alternative computational mechanism but may produce representations whose structure is more naturally described by the formalism in which they were computed. Whether PAM's representations are in fact more interpretable in sheaf-theoretic terms than transformer representations is a testable question that we intend to pursue.

The computational costs are comparable in order to standard attention: the training-time dual form gives $O(T^2Hd)$ per layer, while at inference PAM requires

$O(Hd^2)$ per token using a fixed state of 49,152 floats per layer regardless of sequence length, compared to a KV cache that grows linearly with context. All results reported here are from single training runs on a single RTX 4090, and the 4× throughput gap relative to the transformer (23k vs. 96k tokens/second) reflects the arithmetic cost of complex linear maps and the absence of custom CUDA kernels rather than a fundamental asymmetry in computational complexity.

CONCLUSION

In this work we constructed a language model that operates entirely in complex-valued Hilbert space and evaluated it against a matched transformer on WikiText-103 at ∼100M parameters. Our principal findings are the following:

1. Phase-Associative Memory, which accumulates associations in a complex matrix state via outer products and retrieves via the conjugate inner product, reaches validation perplexity 30.0 on WikiText-103, within ∼10% of a matched transformer (27.1) trained under identical conditions, despite 4× arithmetic overhead from complex computation and no custom CUDA kernels.

2. Holographic binding in vector-state models fails due to the $O(1/\sqrt{n})$ capacity degradation of superposed associations. Upgrading the state from $\mathbb{C}^d$ to $\mathbb{C}^{d\times d}$ resolves this, consistent with the classical analyses of superposition capacity in holographic reduced representations [80].

3. Both magnitude and phase must be free to vary for the conjugate inner product to function as a retrieval mechanism; normalizing queries and keys to unit magnitude causes generation to collapse into repetition.

4. Supplementing PAM with sparse windowed attention provides no benefit at this scale, suggesting that the information attention would contribute is already being captured by the complex recurrence.

5. PAM's retrieval operation implements the same mathematical structure as the quantum semantic framework [48]: outer-product accumulation of associations in a complex Hilbert space and projection of a query onto the accumulated state through the conjugate inner product, the operation that computes measurement probabilities in quantum mechanics.

Whether the ∼10% gap to the transformer narrows or widens at larger scales, whether it is attributable to complex-valued computation rather than the matrix-state structure alone, and whether PAM produces a different contextuality profile than transformer-based models when probed with the CHSH protocol [66] are questions that the experiments currently underway are designed to address.

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